

A research rail diorama seeking its next chapter



Developed at the National Research Council Canada by the Digital Technologies Research Centre, in collaboration with Design and Fabrication Services, this platform combines an operational HO railway, relocation-ready modular construction and museum-quality Canadian-inspired scenery.



WHY IT STANDS OUT

- Approximately 4.53 m × 2.54 m, with all four seasons and several Canadian-inspired environments.
- Ottawa-inspired station, Montréal-style urban district, port and rail yard, Prairie scene, mountain, tunnels and ski area.
- Extensive 3D-printed, hand-painted and weathered details, with numerous figures and narrative scenes.
- A hands-on development and learning platform for multiple undergraduate students.

RAILWAY SYSTEM

- Digitrax DCC with VIA Rail and CN locomotives, both with lighting and sound.
- Two trains can operate simultaneously on a two-level route with bridges, tunnels, passing tracks and a classification yard.
- Turnouts and Tortoise switch motors are installed; DCC turnout-control programming remains to be completed.

TRANSFER PACKAGE

- Approximately 30 numbered modules with matching wooden transport crates.
- Custom two-section table: ten wheeled legs, three additional centre-span supports and module levelling points.
- DCC controller, locomotives, passenger and freight rolling stock, including 3D-printed containers.
- Designed to be dismantled, transported and reassembled; individual modules fit through standard openings.

CURRENT CONDITION

- Transferred as-is. No maintenance, cleaning, repairs or functional testing will be performed before transfer.
- Last active development period: 2022. Plan for inspection, track cleaning and commissioning after reassembly.
- No computers, networking equipment or custom experimental sensors are included. The recipient must manage dismantling, transport and reassembly independently.

AN OPPORTUNITY FOR REUSE

Museums • academia • science outreach • model railway organizations • research prototyping

Photos, history and details: patrickpaul-perso.github.io/dt-diorama/

